

- 7 (a) moving magnet gives rise to/causes/induces e.m.f./current in solenoid/coil B1
 (induced current) creates field/flux in solenoid that opposes (motion of) magnet B1
 work is done/energy is needed to move magnet (into solenoid) B1
 (induced) current gives heating effect (in resistor) which comes from the work done B1 [4]
- (b) current in primary coil give rise to (magnetic) flux/field B1
 (magnetic) flux / field (in core) is in phase with current (in primary coil) B1
 (magnetic) flux threads/links/cuts secondary coil inducing e.m.f. in secondary coil B1
*(there **must** be a mention of secondary coil)*
 e.m.f. induced proportional to rate of change/cutting of flux/field so not in phase B1 [4]